

MH1301 Discrete Mathematics

Handout 4: Recurrence Relations

For Handouts 4, 5, and 6, we cover Part 2 of this course on Recurrence Relations. We introduce recurrence relations and practise finding recurrence relations of some examples. We leave the part on solving recurrence relations to the next two handouts.

- Introduction and Applications
- Modeling with Recurrence Relations.

Motivation

Here are two examples that we will learn to solve using recurrence relations.

Example 2 (Section 8.1)

What is the minimum number of steps needed to move n disks from one bar to another in the Tower of Hanoi?

Rules:

1. Disks are to be moved one at a time from one bar to another.
2. A disk is never placed on top of a smaller disk.

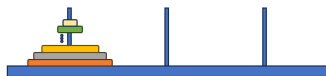
Example 3 (Section 8.1)

How many 0-1 bit strings are there of length n that do not have two consecutive 0's?

Example - Tower of Hanoi

Example 2 (Section 8.1)

What is the minimum number of steps needed to move n disks from one bar to another in the Tower of Hanoi?



Let H_n denote the minimum number of steps needed to move n disks from one bar to another.



H_{n-1}



1



H_{n-1}

$$\begin{aligned} H_n &= H_{n-1} + 1 + H_{n-1} \\ &= 2H_{n-1} + 1 \end{aligned}$$

Example - Tower of Hanoi

Example 2 (Section 8.1)

What is the minimum number of steps needed to move n disks from one bar to another in the Tower of Hanoi?

Solution: Let H_n denote the minimum number of steps needed to move n disks from one bar to another.

- Initial case: $H_1 = 1$
→ same as induction base case
- Recurrence: To compute H_n :
 - Move the top $n - 1$ disks to the middle bar (H_{n-1} moves).
 - Move the largest disk to the right bar (1 move).
 - Move all $n - 1$ disks in the middle bar to the right bar (H_{n-1} moves).
- Hence we get

$$H_n = 2H_{n-1} + 1.$$

Example - Tower of Hanoi

How do we solve the recurrence $H_1 = 1$, $H_n = 2H_{n-1} + 1$?

Intuition:

n	1	2	3	4	5	6	7	8
H_n	1	3	7	15	31	63	127	255

... $2^n - 1$
 $H_2 = 2H_1 + 1$
 $= 3 = 2^2 - 1$
 $H_3 = 2H_2 + 1$
 $= 2 \times 3 + 1$
 $= 7 = 2^3 - 1$

Solution: We prove that $H_n = 2^n - 1$.

- Base step: when $n = 1$, $H_1 = 2^1 - 1 = 1$ is true.
- Induction step: assume that the claim is true for all $n \leq k$, when $n = k + 1$. Then

$$\begin{aligned} H_n &= 2H_{n-1} + 1 \\ &= 2(2^{n-1} - 1) + 1 \\ &= 2^n - 1. \end{aligned}$$

So it takes $2^n - 1$ moves to move n disks from one bar to another.

Reve's Puzzle

Question

What if there are $k \geq 4$ bars, instead of 3, in the Tower of Hanoi?
Does the problem become easier?

The problem (finding a formula for the smallest number of steps) is called **Reve's puzzle**.

The case of $k = 4$ has been solved in 2014.

The solution for larger k is still **unknown**!

Idea of Recurrence

For a given problem with size n , relate the solution of the current problem to those with **size smaller than n** (e.g. $n - 1$, $n - 2$, etc).

That is, construct a recurrence relation in terms of the size of the problem.

- e.g. in the Tower of Hanoi, the recurrence relation is $H_n = 2H_{n-1} + 1$

↙
in terms
of smaller
numbers

Example - Compound Interest

Suppose a person deposits \$10,000 into an account at a bank with interest rate 11% per year with interest compounded annually. How much will be in the account after 30 years?

Solution: Let P_n denote the amount in the bank after n years.

- The initial case is $P_0 = 10,000$.
- Recurrence relation: $P_n = P_{n-1} + 0.11 \cdot P_{n-1} = 1.11 \cdot P_{n-1}$.

We use an iterative approach to find P_n :

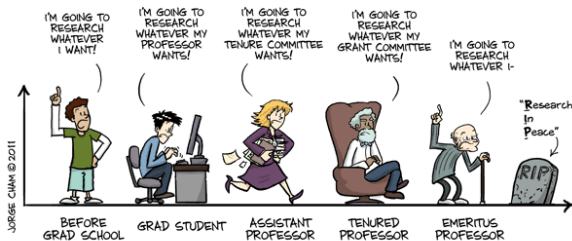
- $P_1 = 1.11 \cdot P_0$
- $P_2 = 1.11 \cdot P_1 = (1.11)^2 \cdot P_0$
- $\vdots$
- $P_n = 1.11 \cdot P_{n-1} = (1.11)^n \cdot P_0$

When $n = 30$, we have $P_{30} = (1.11)^{30} \cdot 10,000 = 228,922.97$.

Example - Graduate Student Job Problem

Every year, each professor graduates 1 student who goes on to become a professor, except first year professors, who produce 0 students. Assume that there are no retirements and professors live forever. If there is 1 professor hired in year 1 , how many professors will there be in year n ?

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Example - Graduate Student Job Problem

Solution: Let f_n be the number of professors in year n .

$$f_0 = 0, \quad f_1 = 1, \quad f_2 = 1$$

$$f_3 = 2, \quad f_4 = 3, \quad f_5 = 5$$

Recurrence: f_n is composed of two parts:

$$f_n = f_{n-1} + f_{n-2}$$

- what we already had last year
- new professors generated this year

The sequence (f_n) is called **Fibonacci numbers**.

Fibonacci Numbers

In the previous examples, the recurrence relations are simpler and we were able to solve them without introducing any new mathematical concepts. How about this recurrence relation for the Fibonacci Numbers?

How do we solve the recurrence $f_0 = 0$, $f_1 = 1$, $f_n = F_{n-1} + f_{n-2}$?

0, 1, 1, 2, 3, 5, 8, 13, 21, ...
** 2 initial values*
We will learn in handout 5 that the following is the solution.

Answer. (See Handout 5.)

$$f_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}, \text{ where } \varphi = \frac{1+\sqrt{5}}{2} \text{ and } \psi = \frac{1-\sqrt{5}}{2}.$$

Example - Strings

1, 2, 3, 5, 8, 13

Example 3 (Section 8.1)

Find a recurrence relation and give initial conditions for the number of 0-1 bit strings of length n that do not contain two consecutive 0's.

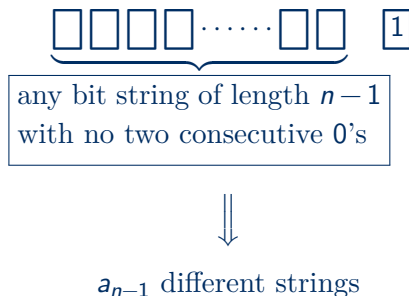
Solution: Let a_n denote the number of strings of length n that do not have two consecutive 0's.

- Initial case: $a_1 = 2$, $a_2 = 3$.

$n=0$ default: 1
 $n=1$ 0
1
 $n=2$ ~~00~~
01
10
11
 $n=3$ ~~000~~
~~001~~
010
011
~~100~~
101
110
111

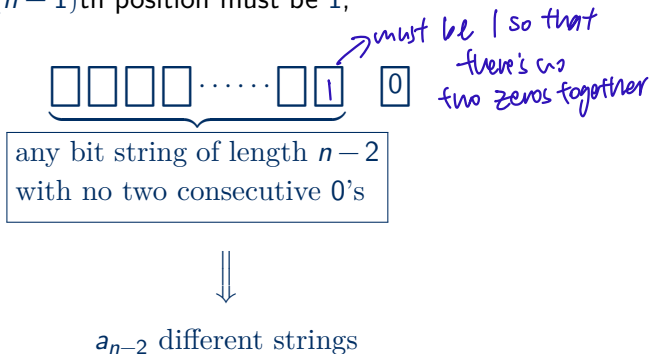
Example - Strings

- Recurrence: consider the last bit of the string.
- If it is 1,



Example - Strings

- Recurrence: consider the last bit of the string.
- If it is 0, the $(n-1)$ th position must be 1,



Example - Strings

Recurrence: consider the last bit of the string.

- If it is 1, there are a_{n-1} different strings.
- If it is 0, there are a_{n-2} different strings.

Hence, we have

$$a_n = a_{n-1} + a_{n-2}.$$

Hence, (a_n) satisfies the same recurrence relation as the Fibonacci sequence (but have different initial conditions).

(Solution of a_n deferred to Handout 5.)

Example - Codewords

Example 4 (Section 8.1)

A computer system considers a string of decimal digits a valid codeword if it contains an even number of 0 digits.

For instance,

- 12301680 is valid
- 13053087013 is invalid

10 possible digits
for each position

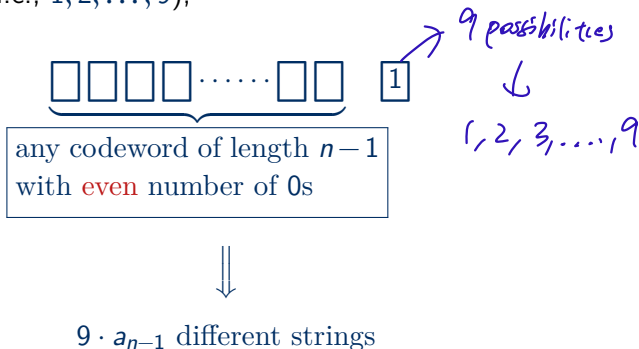
Let a_n be the number of valid n -digit codewords. Find a recurrence relation for a_n .

$a_0: 1$
 $a_1 = 9$ → cannot be '0'.
 $a_2 = 92 \rightarrow (10^2 - 8 \times 1)$
↓
need remove, 01, 02,
..., 09 and 10, 20,
..., 90

Example - Codewords

Solution: Recurrence: consider the last digit of the codeword

- If it is not 0 (i.e., $1, 2, \dots, 9$),



Example - Codewords

Recurrence: consider the last digit of the codeword

- If it is 0,



any codeword of length $n-1$
with **odd number of 0s**

adding both valid
and invalid
parentheses
possibilities

→ can write it as
the code words we
do not want

Total: 10^{n-1}

even # of zeros: a_{n-1}

odd # of zeros: $10^{n-1} - a_{n-1}$



$10^{n-1} - a_{n-1}$ different strings

Example - Codewords

Recurrence: Consider the last digit of the codeword

- if it is not 0, there are $9a_{n-1}$ different codewords
- if it is 0, there are $10^{n-1} - a_{n-1}$ different codewords

good strategy,
but does not
always work

Hence we have

$$a_n = 9a_{n-1} + (10^{n-1} - a_{n-1}) = 8a_{n-1} + 10^{n-1}.$$

(Solution of a_n deferred to Handout 5.)

Reading Assignment

Rosen (7th edition):

- Chapter 8.1 Applications of Recurrence Relations